

KKBOX Churn Prediction

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Predicting Churn Before It Happens

A subscriber's last few actions say more than they realize. We taught a model to listen.

The question

For every paid KKBox subscriber, predict the probability they will not renew once their subscription expires.

Why it matters

Keeping a subscriber is far cheaper than winning a new one. A good prediction is the first step to acting in time.

Our approach

Turn raw subscription history into a per-user snapshot, then let the machine learning models score churn risk.

Source: KKBox Churn Prediction Challenge, Kaggle 2017

Defining Churn Is the Hard Part

Most datasets never say what churn means. A subscription service lets us define it cleanly.

The labeling problem

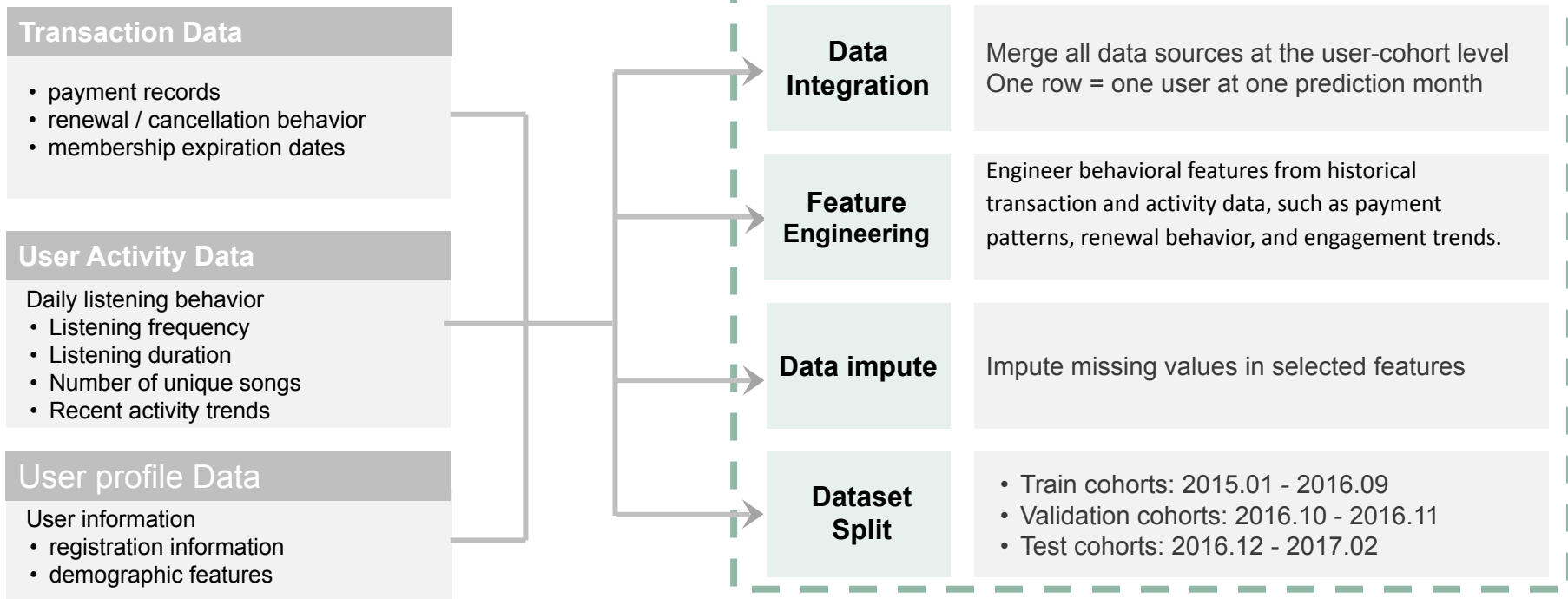
In most data sets, a quiet customer is ambiguous. Someone who buys once and goes silent for a year may or may not have churned.

A subscription model removes the guesswork for defining churn.

How we label churn

- 1 25 monthly cohorts** Each cohort is defined by a cutoff date, the first of that month.
- 2 Pick the targets** Users whose subscription expires in that cohort month.
- 3 Label the outcome** Churned if they don't renew within 30 days of expiration.

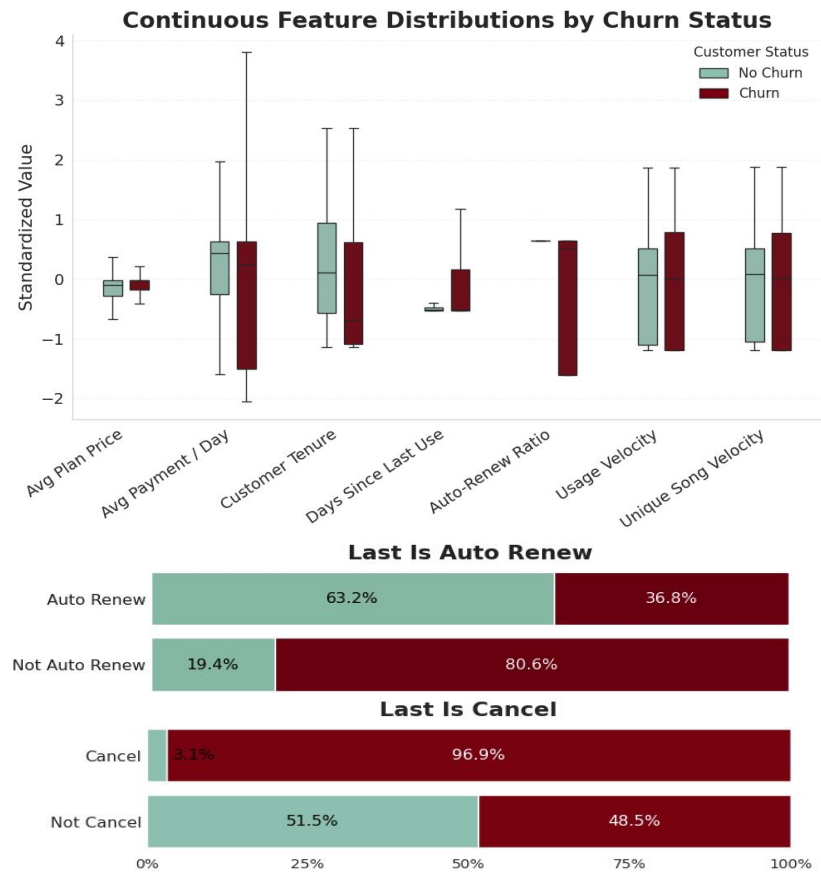
Data Processing Pipeline



① For each cohort, all features are constructed using only historical information before each prediction month to avoid data leakage.

② Generate user-level behavioral features from transaction and activity history, including payment patterns, renewal behavior, and engagement trends.

Feature Visualization



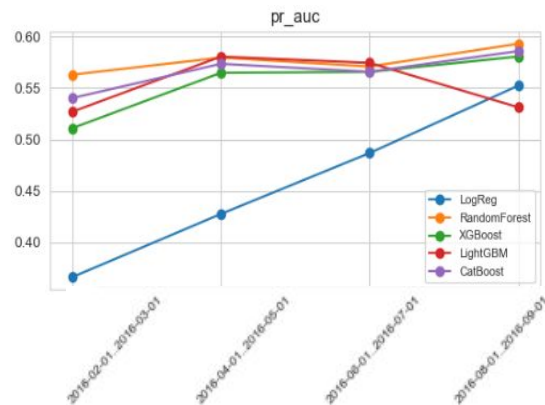
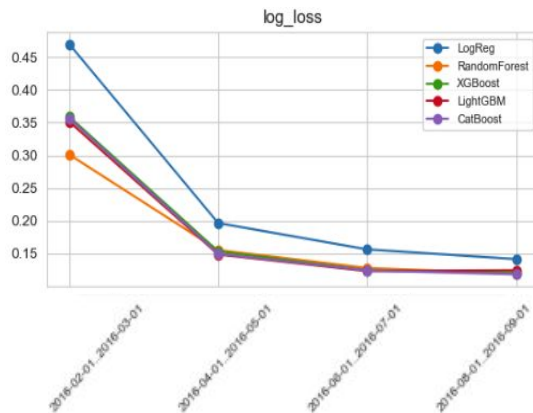
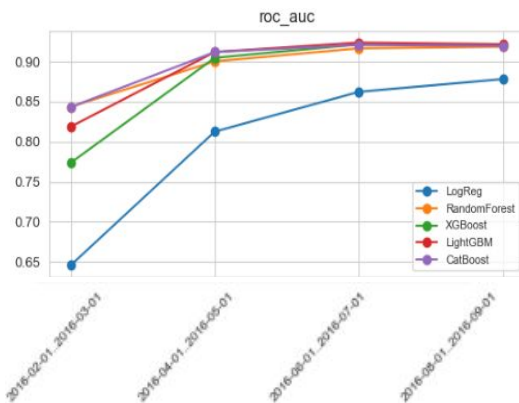
The final feature set contains **9** behavioral variables capturing users' subscription patterns and engagement trends.

Feature summary		
avg_plan_price	days_since_first_trans	days_since_last_use
avg_payment_per_day	num_unq_velocity	total_sec_velocity
ratio_auto_renew	last_is_auto_renew	last_is_cancel

Findings: The selected features show different patterns between churners and non-churners, indicating that they provide informative signals for churn prediction.

Five Models, One Fair Fight

Same weighting, same splits, compared on PR-AUC. XGBoost wins.



model	roc_auc	log_loss	pr_auc
XGBoost	0.940	0.108	0.759
CatBoost	0.938	0.109	0.758
Random Forest	0.938	0.110	0.758
LightGBM	0.940	0.109	0.756
LogReg	0.904	0.135	0.703

We compared 5 models:
XGBoost, LightGBM, CatBoost, RandomForest, LogisticRegression
 using time series cross validation on the 20 train cohorts: Feb 2015 - Sep 2016.

Then training on the whole 20 train cohorts and testing on the two validation cohorts results.

Hyperparameter Tuning

Use Bayesian optimization, learning from past trials to propose promising parameters

Best val PR-AUC from tuning:
 0.7600

Can We Do Better Than One Model?

Barely. And not by enough to give up explainability.

The ensemble we built

- All five models combined, each weighted by how it scored on cross-validation. The val set never influenced the weights.
- The four tree models split the weight almost evenly. Logistic Regression earned almost no say.

0.251	0.2089	0.2013	0.2984	0.0404
CatBoost	XGBoost	LightGBM	RandomForest	LogReg

We shipped XGBoost anyway

A gain this small, measured on three cohorts, does not justify losing per-user SHAP explanations. Explaining an ensemble needs a model-agnostic method that is orders of magnitude slower.

Ensemble vs XGBoost, on val

Metric	Ensemble	XGBoost
PR-AUC <i>better</i>	0.7606	0.7588
Log Loss <i>worse</i>	0.1083	0.1084
ROC-AUC <i>worse</i>	0.9395	0.9402

Tested Once, on Three Unseen Months

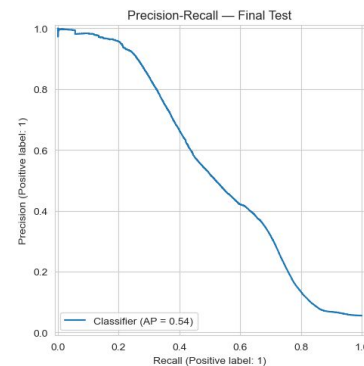
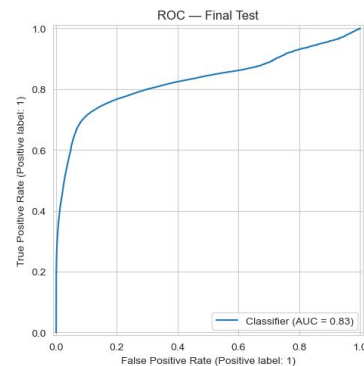
The tuned XGBoost, trained on all 22 train and validation cohorts, then scored once on Dec 2016 through Feb 2017.

Cohort	Churn rate	ROC-AUC	Log Loss	PR-AUC
Dec 2016	8.06%	0.788	0.204	0.624
Jan 2017	4.49%	0.879	0.121	0.475
Feb 2017	3.94%	0.858	0.119	0.379
Overall	5.51%	0.830	0.148	0.542

0.542

overall PR-AUC

on months the model never saw



Overall is computed on the pooled test set, not the mean of the three cohorts.

ROC and precision-recall on the pooled test set.

Nearly 10x Better Than Guessing

PR-AUC 0.542 against a 5.5% base rate. Tuned on validation, test opened once.

0.055

Random guessing. Only 5.5% of test users churn, so that is all chance can give you.

0.542

Our model, on three held-out cohorts it had never seen. A 9.8x lift.

Why PR-AUC, not ROC-AUC

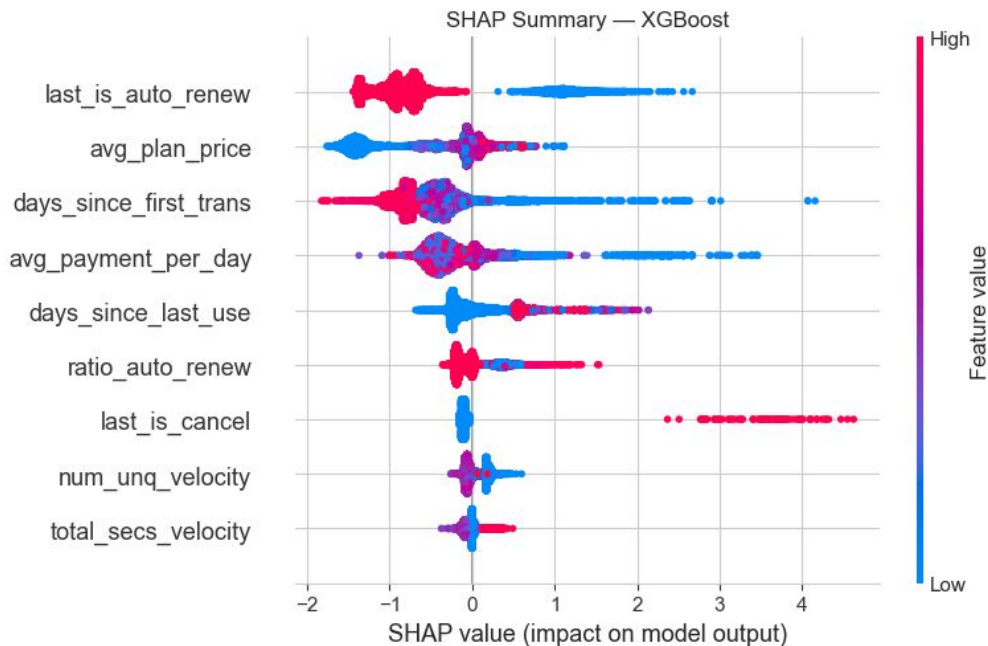
At a 5.5% base rate ROC-AUC flatters: a model can post an impressive number while missing most of the churners. PR-AUC asks what the retention team actually cares about.

The problem got harder over time

Churn fell from 7.96% in training to 6.52% in validation to 5.51% in test. Fewer churners to find means a lower ceiling on PR-AUC. The 9.8x lift already accounts for that.

What Actually Drives Churn

Auto-renew status moves every prediction. A cancellation moves few, but moves them hardest.



How to read SHAP summary

- last_is_auto_renew has the largest average impact. It applies to everyone, and a passive renewal pulls risk down.
- last_is_cancel is rare, so its absolute average looks small. But for the few users it touches, it is the single hardest push toward churn.
- Average impact and peak impact are different questions. SHAP answers both, because it is computed per user.

What SHAP does and doesn't say

It measures what the model weighted, not what causes churn. We report attribution, and we say so.

SHAP Decides. LLM Translates.

Every fact comes from the model. The LLM gives it words.



The Guardrail

The LLM receives only what SHAP surfaces for the user in question: which features mattered, the features values, and which direction the model weighted them. The LLM does not invent the reasons, but rather verbalizes the the structured evidence naturally, so a human can check the narrative against the model.

A Limitation

SHAP is additive, so it reports each factor's marginal contribution. This layer cannot parse conditional interactions. Heavier computation would be needed.

What a retention agent reads

"The model scores this user at 94% churn risk. It weighted their recent cancellation most heavily, alongside a long gap since their last listen. Their auto-renew history pulled the estimate down slightly. Worth trying: a win-back offer before the subscription lapses."

No statistics. No jargon. No invented reasons.

Limitations & Next Steps

Many users with multiple transactions in a day

A better data tracking scheme should be implemented to properly order all transactions

High missingness or lack of specificity in member data

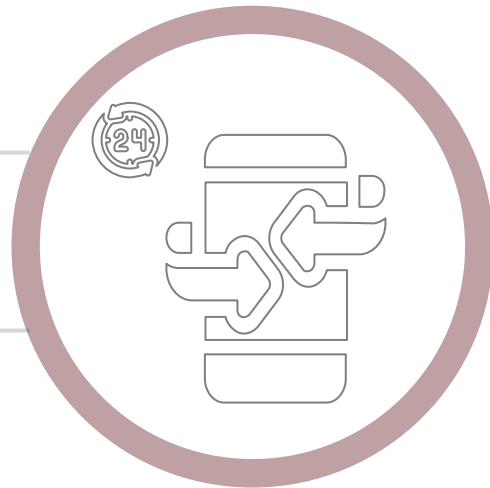
Having better member information could lead to more features to predict churn

Users in multiple cohorts

Might cause model to perform more optimistic than it will in deployment seeing completely new users

Features `avg_payment_price`, `avg_payment_per_day`, and `usage velocities` give mixed/unintended signals

Cleaning up these features and separating the signals might lead to better and more explainable performance



FROM SCORE TO ACTION

Who uses it, how, and what it's worth

- Losing paid users can directly hurt revenue. A company can identify at-risk users early and offer incentives, support, and personalized content to retain them.
- By reducing the churn rate, a company can lower the cost of acquiring replacement customers and build stronger customer loyalty, to create more predictable revenue.
- Churn patterns can reveal problems with a company's products, services, or pricing, helping it develop more effective marketing strategies.



Thank you!

